Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years building **performant frontend systems** using **TypeScript and React**. Experience developing intuitive interfaces for data-intensive applications, with strong foundations in algorithms, data structures, and Linux environments.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Data Structures, Database Systems, Systems Programming

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4.0*
Coursework: Data Structures, Algorithms, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*

- Built **performant React frontend dashboards** integrated with FastAPI backend, optimizing rendering and query patterns to reduce latency from **1.2s to 350ms** for 500+ users accessing data-intensive views.
- Collaborated with **cross-functional teams** to understand user needs and translate them into effective UI solutions, participating in code reviews to uphold coding standards and design philosophy.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*

- Developed **intuitive React interfaces** with TypeScript for a grading platform, implementing responsive UI components that handled 10K+ daily interactions while focusing on UX and user experience.
- Wrote **clean, well-tested code** under guidance of senior engineers, participating in code reviews to learn and uphold coding standards while collaborating closely with teammates on feature development.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*

- Built **data processing pipelines** using Python on Linux, implementing algorithms that improved model performance by 18% through systematic analysis and problem-solving across 100K+ records.
- Demonstrated **strong analytical skills** by designing efficient data structures and algorithms for trajectory planning, working collaboratively with researchers in a fast-paced environment.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*

- Engineered **scalable backend systems on Linux** using Python, implementing algorithms that improved response times by 30% and pipeline efficiency by **68%** through optimization and testing.
- Built **internal tools to increase efficiency** of data processing and validation workflows, collaborating in Agile sprints with strong communication across technical teams.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*

- Developed **React frontend with real-time WebSocket updates** for a CRM platform, implementing intuitive trading-style interfaces for resource allocation and data visualization across 2M+ records.
- Built **full-stack features using MERN stack**, demonstrating ability to adapt quickly in a fast-paced environment while maintaining code quality and clear documentation.

Skills

Frontend: TypeScript, JavaScript, React, Next.js, HTML, CSS, TailwindCSS, responsive UI
Backend: Python, Node.js, Flask, FastAPI, REST APIs, PostgreSQL, MongoDB
Systems: Linux, Git, Docker, algorithms, data structures, systems programming basics
Practices: Code reviews, testing, Agile/Scrum, collaborative development, UX focus

Projects

FinNexus Live | *Next.js, TypeScript, Plaid, MongoDB, WebSockets*

- Built a **performant fintech platform** supporting 50K+ users and 1M+ transactions with real-time data visualization, implementing intuitive TypeScript/React interfaces for financial data and transaction management.

PiXMod Live | *Next.js, TypeScript, MongoDB, Stripe, TailwindCSS*

- Built an **intuitive SaaS interface** with Next.js and TypeScript, focusing on UX best practices and responsive design for image processing features across 100K+ user interactions.

CPU-Scheduling Simulator Live | *C++, JavaScript, React, Algorithms*

- Implemented **scheduling algorithms** (FCFS, SJF, SRT, Round Robin, MLQ) demonstrating strong understanding of systems programming, data structures, and algorithm design with 25% TAT reduction.